

IoT Emergency Health Monitoring System

Real-time Vital Tracking with Automated SMS/Call Trigger & Location Transmission

MICROCONTROLLER
ESP32 (30-Pin)

SENSORS
MAX30102 / NEO-6M

CONNECTIVITY
SIM800L GSM/GPRS

DISPLAY
0.96" OLED (I2C)

1. Project Overview & Abstract

This project presents a portable, IoT-enabled wearable emergency health monitoring device designed for real-time monitoring of critical human physiological parameters. Driven by an ESP32 microcontroller, the system measures heart rate (BPM) and blood oxygen saturation (%SpO2) via the MAX30102 pulse oximetry sensor, while continuously displaying time, connectivity status, and vital stats on an OLED screen.

Emergency Response Mechanism

The system triggers automated alerts under two conditions: (1) Manual press of the Emergency SOS button, or (2) Automatic detection of life-threatening anomalies (e.g., severe bradycardia/tachycardia or hypoxemia). Upon trigger, it obtains real-time GPS coordinates via NEO-6M and initiates an automatic SMS with a Google Maps link and a direct voice call via the SIM800L module.

2. Hardware Modules & Architecture

ESP32 WROOM-32

Serves as the main control unit, featuring dual-core 240MHz processing, integrated Wi-Fi/Bluetooth, and multiple hardware UART/I2C buses to manage sensor sampling, display updating, and cellular/GPS communications.

MAX30102 Pulse Oximeter

Optical sensor combining red and infrared LEDs with photodetectors to measure PPG signals for calculating heart rate and oxygen saturation (SpO2), utilizing I2C communication (0x57 address).

NEO-6M GPS Module

Retrieves NMEA sentence strings from satellite constellations to determine precise geographic latitude and longitude coordinates. Interfaced via hardware UART2.

SIM800L GSM Module

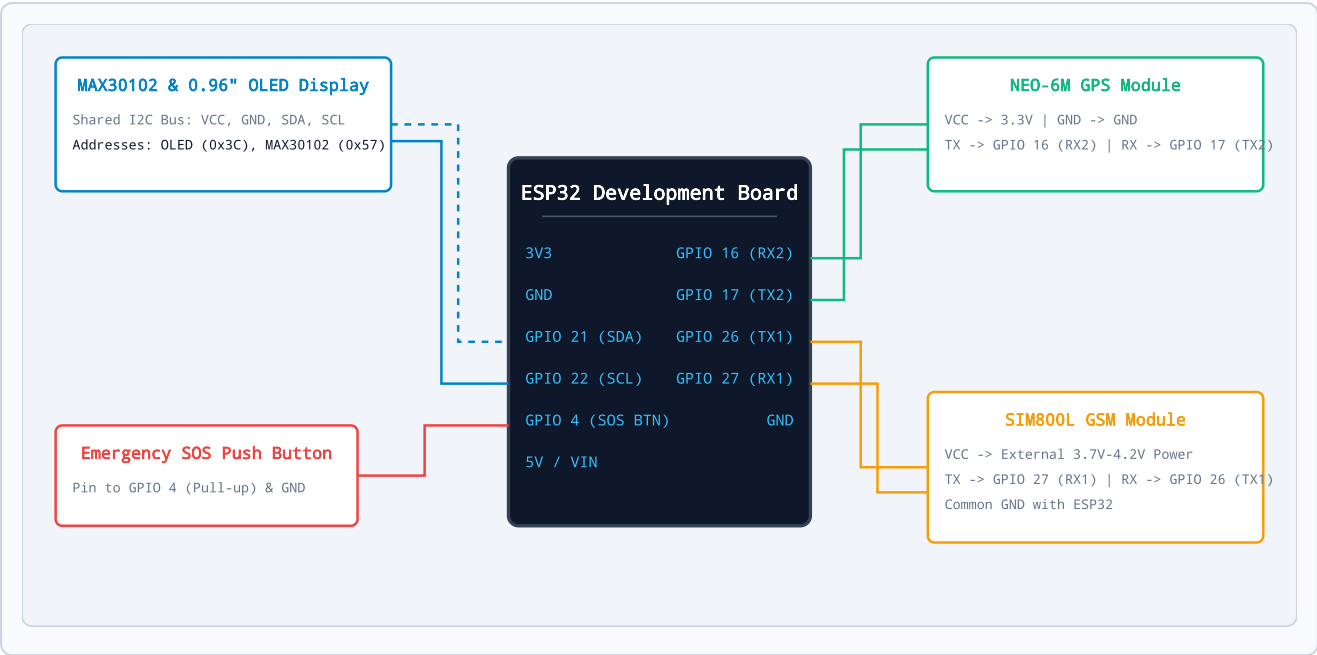
Quad-band GSM/GPRS module executing AT commands via UART1 to send SMS text messages containing live Google Maps URLs and place voice phone calls during emergency states.

3. System Specifications & Thresholds

Parameter	Normal Range	Emergency Threshold	Action Taken
Heart Rate (BPM)	60 – 100 BPM	< 40 BPM or > 130 BPM	Auto-Trigger Call & Location SMS
Blood Oxygen (SpO2)	95% – 100%	< 90% (Hypoxia Alert)	Auto-Trigger Call & Location SMS
Manual SOS Button	Idle / Open	Pressed (GND Pulled Low)	Immediate Call & Location SMS

4. System Circuit Diagram & Interconnections

Below is the electrical schematic mapping the interconnections between the ESP32 microcontroller, sensors, display, push button triggers, and communication modules.



5. Operational Workflow & Program Logic

1. Initialization & Baseline Monitoring
- Upon boot, the ESP32 initializes I2C and UART communications. It establishes connection with Wi-Fi/NTP for time sync (displayed on OLED) and continuously samples infrared/red light attenuation from the MAX30102 sensor.
2. Real-Time Signal Processing
- Photoplethysmogram (PPG) data is filtered to extract pulse rate (BPM) and oxygen concentration (%SpO2). The results are continuously formatted and refreshed on the 0.96" OLED display.
3. Critical Condition / Emergency Trigger Detection
- The microcontroller continuously evaluates two emergency conditions:
- **Manual Trigger:** Pressing the physical tactile push button attached to GPIO 4.
 - **Automated Metric Trigger:** Heart rate dropping below 40 BPM or exceeding 130 BPM, or SpO2 dropping below 90%.
4. Location Fetch & Dispatch Execution
- Once triggered, the ESP32 reads the latest valid NMEA sentence from the NEO-6M GPS module, constructs a standard Google Maps URL (e.g., <http://maps.google.com/maps?q=LAT,LON>), displays "Sending Location..." on OLED, commands the SIM800L via AT commands to send an emergency SMS, and immediately initiates a voice call to the predefined caregiver number.

6. Bill of Materials (BOM) Pin Connection Summary

Component	Pin Name	ESP32 Pin Connection	Notes
MAX30102 & OLED	SDA / SCL	GPIO 21 / GPIO 22	Parallel I2C Bus with pull-ups
	VCC / GND	3.3V / GND	Main 3.3V power rail
NEO-6M GPS	TX / RX	GPIO 16 (RX2) / GPIO 17 (TX2)	Hardware Serial 2
	VCC / GND	3.3V / GND	Regulated 3.3V input
SIM800L GSM	TX / RX	GPIO 27 (RX1) / GPIO 26 (TX1)	Hardware Serial 1
	VCC / GND	External 3.7V - 4.2V / GND	Requires high peak current source (>2A)